

MICE in the Machine: Creating a computational model of novelist techniques.

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Abstract

This paper focuses on the novelist technique known as the MICE quotient, a way of reducing stories to their individual story threads. MICE does this reduction by decomposing a story into four categories: Milieu, Idea, Character, and Event. Each category has its own story threads, which are arcs that follow a format for how to begin and how to conclude. Additionally, by following the MICE thread format, authors can inspect how each thread impacts the reader's or player's expectations of the storyline; by understanding these expectations, authors can focus on the story they are telling. Interestingly, the structure that MICE uses maps well to a behavior tree. Thus, this paper focuses on building a computational model for generating stories during gameplay. Our MICE model uses a reactive planner at its core to switch between threads based on player input. As we built a proof of concept, however, we discovered issues and deficiencies of a strict one-to-one mapping of the MICE technique to a computational model, which we discuss in this paper. We also discuss possible ways to overcome these issues and next steps for a more comprehensive story generation tool with MICE.

Introduction

Interactive Stories are often burdensome to write because of the expansive nature of diverging story states and their complexities. Therefore, authors must ensure all narrative paths make sense regardless of how a player reaches a particular point within a game's story. If the author cannot keep the story consistent between two or more points, then a player's sense of immersion or satisfaction may be disrupted, leading to a lessened experience. This increase in story complexity is often referred to as authorial burden (Méndez et al. 2019; Ryan, Mateas, and Wardrip-Fruin 2015).

To ease the authorial burden within interactive story design, researchers have developed AI systems that focus on reducing the story complexity space by offloading parts of

story generation to an AI system, such as Mimi'sbrunnur (Stefnissón and Thue 2018) or drama managers (Chen, Nelson, and Mateas 2009), or scene-based story generation, like Lume (Mason, Stagg, and Wardrip-Fruin 2019). But unlike most of these systems, the work presented in this paper takes techniques used by novelists to lower the authorial burden through tracking story arcs and the questions they present to the player.

To better understand this approach, we adopted a framing that traditional authors use for novel writing, which is to consider stories as a set of "promises," which take the form of questions, such as, "will the protagonist save the world," and answered within the story, such as, "yes they will save the world." The questions themselves can be anything that the author wishes to turn into a story arc, such as surviving on an island to learning to overcome anxieties. By rearranging stories into a question-and-answer model, we can now see when a story line has failed to answer an open question; often, these open questions frustrate the reader by never giving them closure for a certain plot point. Yet, Interactive Storytelling is more complex than the traditional novel since the story can diverge based on player input. Additionally, when the story diverges, it can miss answering a story question, which could create a plot hole that will frustrate the reader.

As for the question-and-answer framing, Orson Scott Card describes this breakdown and how to categorize all questions into one of four types, this method is commonly referred to as the MICE quotient, or MICE (Card 1988). The MICE model of storytelling reformulates narratives into these four general question categories, calling each new addition a story thread. Then MICE explains how each thread starts a story question and how each thread should resolve, ensuring that the opened story question has been answered. Interestingly, the structure in which MICE keeps track of questions, and this structure seems to map well to

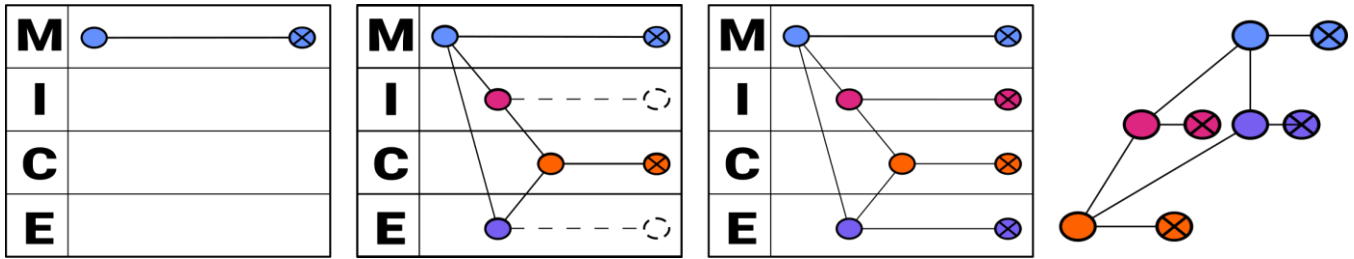


Figure 1 This figure visualizes the story complexity of MICE story graphs. Starting from left to right, the first graph is a single-threaded story, the next is a multi-threaded story that does not close out all threads, and right of that graph is a fully closed graph. The rightmost graph is a mapping of a MICE graph to a behavior ABL-like structure to indicate how well MICE story graphs translate to an ABL tree.

behavior trees. This mapping is one of the key reasons our work uses A Behavior Language, ABL (Mateas and Stern 2002), and why Façade (Mateas and Stern 2005) has a significant influence on this work.

As for this paper, we present an explanation of MICE, a computational implementation of a strict MICE model, the deficiencies that a strict computational MICE model has, and ways to ease and fix the inherent issues within the rigid model. To the best of our knowledge, this is the first work to incorporate MICE into a computational model of storytelling.

Story Promises, Resolutions, and MICE

A story often makes a promise about the tale that it's telling, and Card argues these promises are a contract to the reader, which are often presented as questions to the reader, such as the murder mystery with a victim's discovery; the story should ultimately answer the question of, "who is the murderer?" If the story fails to follow the murder investigation and instead focuses on another story—like a character who was passing by during the story's start and had seen the detective beginning the investigation. If we then follow this character after some abrupt POV shift, then the story may cannot deliver on its promise, assuming that the passerby is not the murderer and holds no point to the murder mystery. Then the story has failed to deliver on its promise. Card states that if a story fails to uphold to its original premise and the original questions, then the story will leave the reader confused and disappointed, since these questions translate to unresolved story lines. Do note, however, that if the story's framing had been focused on the aftermath of a murder and how characters respond to it, then readers can find the story satisfying, even though the question of, "who is the murderer," is never answered, and this is fine to do since the murderer's identity question isn't the one that's been asked; instead, the question would be something akin to, "how do we move on after a death?"

To avoid the failure case of unresolved questions, Card introduces MICE, which focuses on generalizing stories into their promises. MICE does this breakdown by decomposing a story into four different distinct threads: Milieu, Idea,

Character, and Event. These four categories make up the technique's name: MICE. Do note that these threads are categories, not a set of prescribed stories. Below is a description of each category, how they start, and how they resolve; another note, threads are the questions-and-answer set that was mentioned earlier in this paper.

Milieu threads start by a character finding themselves in an unfamiliar location, like Dorothy in "The Wizard of Oz." The Milieu story focuses on if the character can leave the location, like Dorothy using her slippers, or the character comes to terms with never returning to their original location, accepting the new status quo.

Idea threads focus on uncovering a mystery or learning more about unknown elements of the story. The murder mystery is a prime example of an Idea story, since the key focus of these stories are a singular question like, "who is the murderer?" Idea story threads resolve when the question is resolved, such as the detective discovering the murderer's identity and confronting them.

Character threads focus on a character's internal struggle, such as going from one internal state to another. These states often focus on changing emotional states, to a change within personality, such as characters going from arrogant to humble; it is up to the author how they want the character to change. These resolve when the character finds a new stable point, or an inner peace, or the author signals that the character has changed; they can also end with the character failing to change, depending on how the author wants their story to end.

Finally, **Event** threads focus on an external event, such as a natural disaster, that upsets the status quo of the story world. These threads often resolve once the event is over, either by character intervention or the event causes a new status quo that the characters must accept. This is often the driving thread within apocalypse movies.

For encoding MICE into a computational model, the rightmost graph of Figure 1 reorders the complete story graph into a tree-like structure, with a few deviations—such as the horizontal node links within each level of the tree; these crossed out nodes are considered the resolution node, as described on the right side of figure 2. Otherwise, the structure is still the same, a Milieu node begins, spawns

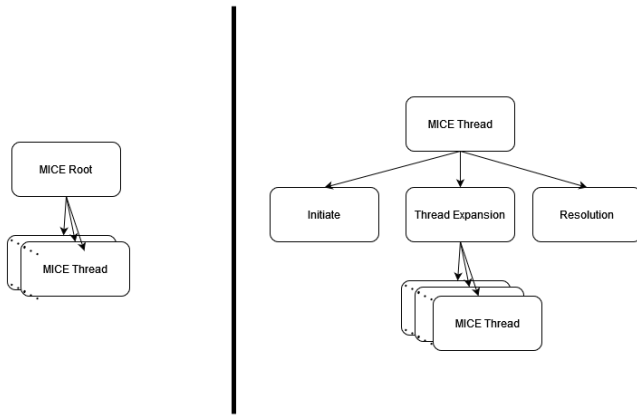


Figure 2 This is the basic model for all MICE threads. This structure is used every time a thread is instantiated. This generic representation allows for the complexities of MICE-based stories to be used in a consistent and simple way.

three nodes, an initial Idea thread node, an initial Event thread node, and the Milieu’s own resolution node, which does not resolve until all its nested storylines resolve first. Therefore, in this case, the lowest level must resolve and be removed from the tree (the Character storyline), then the level above it (the Idea and Event nodes), then finally can the Milieu resolve. Another way to think of this kind of opening and closing is a left to right depth first search on the tree. But do note that at any point, any active story thread can present new material to the player, as in the topmost Milieu node can still impact the story while the tree is in its fully expanded state.

While the beginnings and the resolutions of each thread are well explained within Card’s description, there is still some ill-defined parts of MICE, mostly to do with conflict. A reason for this is that conflict can take many forms and is dependent on the kind of story being told. A Milieu’s conflict can either focus on understanding the political framing of the new world or discovering how magic works within the world; there are numerous ways these storylines can go; it is up to the author how they wish to progress the conflict for each MICE thread.

While single thread stories are fine, sometimes an author wishes to tell a more complex story, such as a character examining an inner turmoil while journeying across a new land, such as how Frodo struggles with the burden of the one ring while traveling to Mount Doom in “The Lord of the Rings.” To better understand story braiding, Figure 1 shows a linear Milieu thread that opens and immediately closes. But combine one or more threads, nest them together, and the story becomes more complex. This complexity does add to the authorial burden, and the author must now ensure each MICE thread closes out properly. But this interlocking of threads is often desirable, for the stories become more unpredictable, which can keep a reader’s interest since they are

unsure of what will happen next. This interlocking technique is also known as story braiding. Additionally, by braiding these threads together, authors can delay a story thread but still keep the story engaging by following another thread.

Figure 1 contains an example of story braiding within the third most graph from the left. Here a Milieu story opens the story followed by an interior story of an Idea and Event thread (each of these story threads can switch between being the main story, creating an interweaving effect). From there, a fourth story thread opens, focusing on a character’s response to everything; any four of these threads can become the main story thread that the player follows. All four of these stories can be the main story at any point and can switch to a different thread at any point. Finally, the four threads will complete their storylines, and the narrative will be over.

Figure 2, on the other hand, focuses on the MICE trees that our system uses. On the left, the initial MICE node is at the topmost position. This initial MICE node now goes into its expansion state, where it can spawn an arbitrary number of children MICE threads. Each child node will spawn its own initiation node, which is effectively a MICE question such as, “who is the murderer.” Each MICE thread node will have its own expansion state. By doing this hierarchical nesting, our system captures nested stories like the Milieu discussed earlier, where a nested Character thread exists within the Milieu story; this nesting allows for story braiding. Finally, each thread will have its own resolution node.

Since the story thread structure of MICE maps well to behavior trees, we decided that ABL, a reactive planning language that uses behavior trees. By using ABL’s behavior trees, our computational model of MICE can switch between story threads, setting some storylines to the background while setting others storylines as the main one; thus keeping track of all the story threads and can remove story threads if the correct story state comes in from the game.

An example of this would be an Idea storyline that becomes inactive for most of the story, but activates when a resolution for the Idea question appears within the story. Our system achieves this effect by encoding the question/answer sets within JSON, with a question structure containing an answer type. Whenever a player progresses the story, possible answer types are sent to the agent, and it checks each active node if that node has been satisfied. If so then remove the node. If not, continue the story until all nodes are satisfied. As of now, all open story nodes are placed on the tree in a static way to better test the system.

Discussion/Future Work

To test how well MICE maps to a computational model, we implemented the model described above, a system that

opens MICE story threads, tracks them through ABL, and resolves them whenever a story condition has been met. We used Unity as a way to interact with the system, using a simple Interactive Fiction prompt built in conjunction with Ink, an interactive narrative language. Our Unity application communicates with a server where the ABL agent is running, sending over relevant story and player data. But as we implemented the ABL agent, we realized that MICE has no built-in way of progressing a story; there is no internal pacing mechanism within Card's description of MICE.

The MICE technique provides story consistency, but MICE offloads all pacing concerns to the author. The threads themselves have no structure other than when a thread is opened and when it should be closed. Because of this, a strict computational model of MICE has much to be desired to fully generate a story on its own; it needs some sort of pacing model. Since this is the case, currently we are looking into using MICE as a mixed-initiative tool, where an author can feed in story states and what MICE threads are open within the story. The MICE tool will then return possible conclusions for the open story questions. We hope to achieve this by encoding multiple generic story questions and answer sets in JSON or some other data format. We also believe that a different language might be more beneficial to this approach since the generation process will not be at a game's runtime, and a logic program like Prolog would be more suited for this kind of story generation.

As for how to include a progression structure within the MICE system, we look to other Interactive Fiction approaches such as modular scenes and drama managers, which could give MICE the knowledge representation and reasoning needed to generate stories. By modularizing the scene, we hope to use these scene components as the building blocks for stories—like how Lume uses its scenes to build up a story. A three-act plot structure through a reactive agent is also being considered. We are currently designing a reactive drama manager, much in the same vein as Façade's drama manager. We also hope to make the pacing system modular, allowing for other pacing technique to be used, such as the Kishōtenketsu story structure—a four act structure that focuses on a twist within the story between acts three and four.

Conclusion

In this paper, we present the MICE quotient, as described by Card, and how to reformulate the MICE technique into a computational model for storytelling. We first explain how MICE handles stories, by breaking them down into four categorical threads, the kind of story questions MICE requires, and how to resolve open story questions; we also explain nesting stories, creating story braids. After this, we showcase the behavior tree-like structure of MICE and how it maps to a behavior tree. Then, after experimenting

with a strict MICE model, we explain the issues with this approach and the possible ways to fix or enhance this approach to create a more robust generative system.

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